THE LAST PROTOCOL
ENGINEERING NOTEBOOK - RECOVERED

The following pages were recovered from the calibration engineer’s private notebook.

1. BLOCK STRUCTURE
The final archive transformation operates on 64-bit blocks.
Each block is divided into two 32-bit halves.
The engineer repeatedly emphasized that the transformation is reversible without constructing a separate inverse function.
Notation: L(r+1) = R(r); R(r+1) = L(r) XOR F(R(r), K(r), r)

2. ROUND MATERIAL
Eight rounds are performed. Every round receives a different 256-bit-derived key value. Only the first eight bytes of each round material are consumed by the round function.
The key is not added directly. Break it into the two words first.
The round function works on a 32-bit word.
Circled constants: 0x9E3779B9 (round), 0x045D9F3B (mixing).

3. OPERATION ORDER
1. XOR the right word with both 32-bit key words
2. Add the round constant multiplied by the round number
3. Rotate left by a round-dependent amount
4. XOR with a shifted copy of itself
5. Multiply by 0x045D9F3B modulo 2^32
6. XOR with itself shifted right by 16
Margin note: The rotation grows by three each round.
Do not rotate right.

4. BYTE ORDER
All multi-byte values in the archive are written in network
byte order. Read the first four bytes as the most significant
word. Read the next four as the least significant word.

5. KNOWN TEST VECTOR
A test block was recorded before the final archive was produced.
Plaintext block: 52 45 42 55 49 4C 44 3A
Ciphertext block: 8A DF 73 64 0B D9 77 3C
If your implementation cannot reproduce this block, your byte
order or round function is wrong.

6. FINAL NOTE
The archive is not the final message.
The first successful decryption only tells you that the
machine has been rebuilt. What it says next is another problem.

END OF NOTEBOOK